Ethnic clusters in trademark filings and patent invention: preliminary results, surname-only

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Brief note testing two questions on the same underlying data layer: (a) within ethnic clusters in a NICE class, is the ΔKL profile of filings systematically different from the cross-class baseline — i.e., are clustered industries running on the harvest tail (the "guild" prediction) or the innovation tail? (b) does the surname-imputed composition of *patent inventors* look similar to the trademark debut composition, and what does that tell us about whether the post-2010s rise in API/Asian filers is a cross-system phenomenon or a trademark-corpus artefact?

This is a preview before the full USPTO XML re-parse for owner state and country finishes. The re-parse (currently running, $\sim$ 30–60 minutes) will let us strip foreign-domiciled trademark filings and recompute the trademark composition for US-domiciled filers only. The patent-side analysis below already uses inventor names directly (PatentsView `g_inventor_disambiguated`, 24M patent-inventor rows, 4.26M unique inventors 1990–2024).

(1) Ethnic clusters in NICE classes: harvest or innovation?

For each (NICE class, surname-imputed ethnicity) cell we compute fractional debut counts via Census 2010 surname-race probabilities (Word, Coleman, Nunziata, & Kominski, 2008), and the n-effective-weighted mean ΔKL on debut filings in that cell. An *enclave* cell is one where the ethnic group's share of individual-name debuts in the class is over 1.5× its overall share, the share is $> 10\%$, and the effective n is > 500.

Enclave ΔKL is on the innovation side, not the harvest side – with one major caveat. N-weighted mean ΔKL in enclave cells versus non-enclave cells within each ethnic group:

Ethnic group	Enclave mean ΔKL	Non-enclave mean ΔKL	Difference
API	+0.088	+0.007	+0.081
Hispanic	+0.083	+0.020	+0.063
White	–	–	no enclave cells
Black	–	–	no enclave cells

Table 1: Enclave cells run *higher* on the innovation tail ($\Delta KL > 0$) than non-enclave cells of the same ethnic group, not lower. The guild prediction (clusters $\rightarrow$ harvest tail) is not supported in our trademark data on first read.

The major caveat is the foreign-applicant confound. Almost every API enclave we find sits in a consumer-manufactured-goods NICE class: Carpets (027), Lighting/Heating (011), Lace/Embroidery (026), Fabrics (024), Household Utensils (021), Furniture (020), Hand Tools (008), Ropes/Nets/Textiles

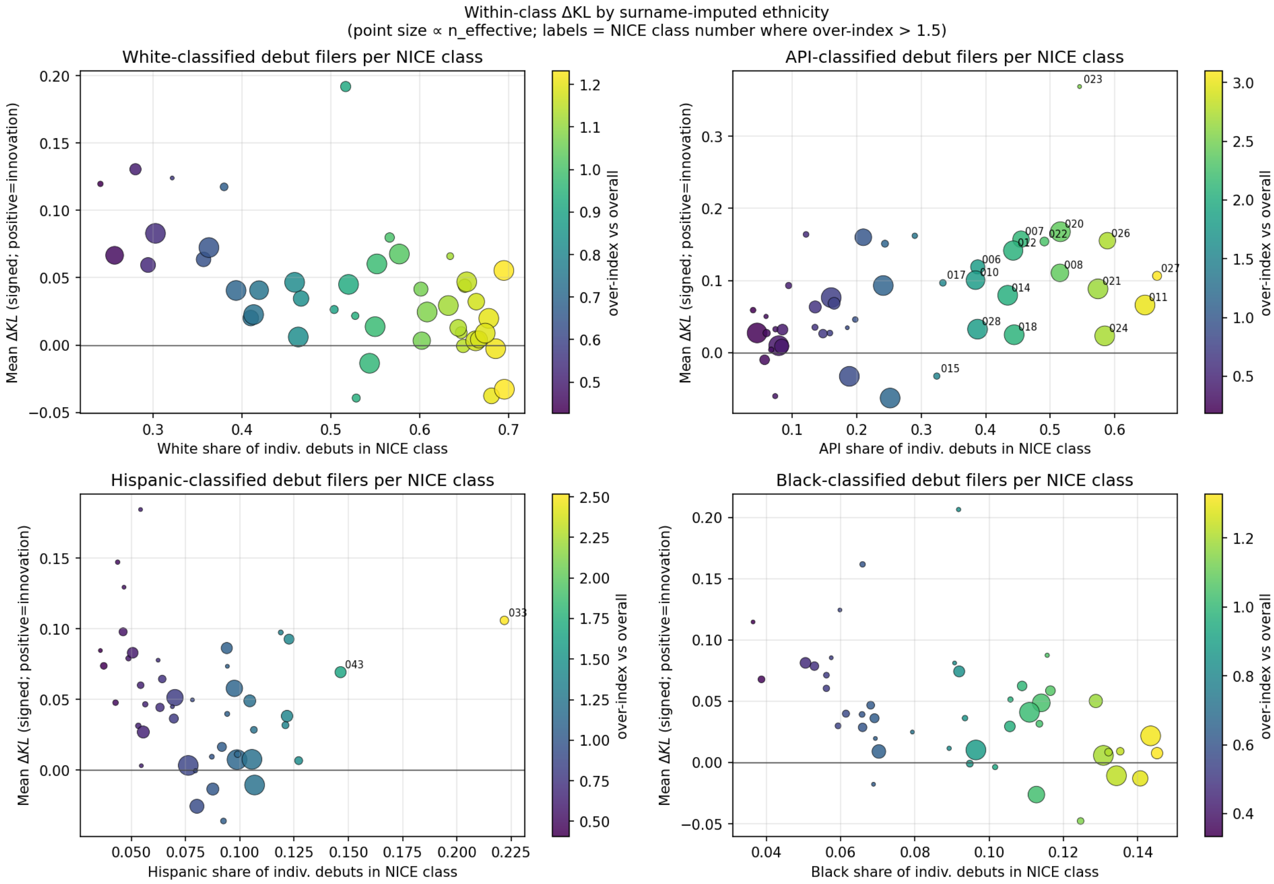

Figure 1: Per-NICE-class mean ΔKL on debut filings, against that ethnic group's share of individual-name debuts in the class. Point size $\propto$ effective n; colour = over-index vs. the ethnic group's overall share of all individual debuts. NICE class numbers are labelled where the over-index exceeds 1.5.

(022), Machines (007), Leather (018), Vehicles (012), Jewelry (014), Common Metals (006), Toys (028), Surgical/Medical (010). The (019 Building, 003 Cosmetics, 025 Clothing) classes also rank high. These are exactly the sectors where the Chinese-applicant filing surge of 2018–2021 concentrated — foreign manufacturers seeking US trademark protection for export brands. The "API enclave $\Delta KL > 0$" result is therefore confounded with "Chinese cross-border applicants writing goods/services prose that is novel against the US baseline." The full re-parse in progress will let us strip foreign-domiciled filings and recompute.

The Hispanic enclaves are the cleaner domestic-pattern test: **Alcoholic Beverages (NICE 033)** and **Hotel/Restaurant/Food (NICE 043)**. Hispanic-classified individuals are 1.5–2.5× over-represented in these classes vs. their overall share. Mean ΔKL in these enclave cells is approximately +0.08, again on the innovation side rather than the harvest side. This contradicts the strong form of the Borjas (1992) / Kerr-Mandorff (2023) ethnic-cluster-as-stagnation prediction in the trademark data. Hispanic-American restaurant/food/beverage entrepreneurs file *novel* goods/services prose at a higher rate than the corpus baseline, not legacy/recycled prose.

What we don't see. No White or Black enclave cells trigger our 1.5× over-representation threshold. White is the majority in essentially every class, so no class is *over-represented* for White by that much. Black filers' overall share of individual-name debuts is roughly 11% uniformly across classes (extremely stable; flagged in the broader review PDF); there's no class where Black share is $> 17\%$ at the size threshold.

(2) Patent inventor ethnic composition, 1990–2024

PatentsView ships disambiguated inventor records: each `inventor_id` appears once per patent they're listed on. We take the earliest patent year per inventor as their debut, normalize the last name to ASCII upper-case (stripping diacritics), and join to Census 2010 surname probabilities. 4,254,889 unique inventors with a usable surname; 3,121,166 (73.4%) match a Census surname.

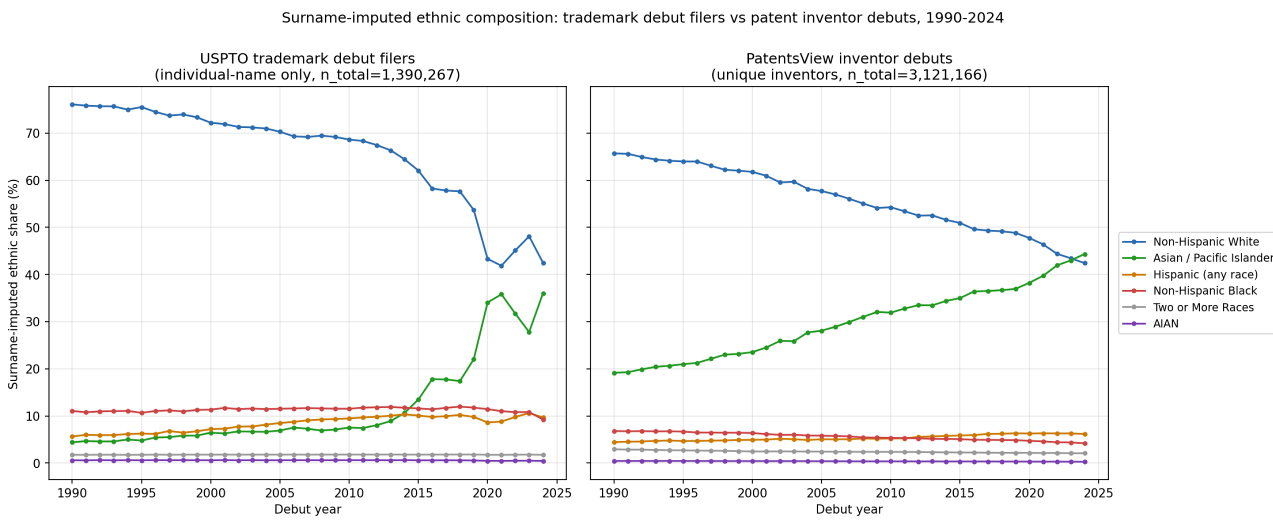

Figure 2: Left panel reproduces the trademark debut-filer composition we already had. Right panel is the patent-inventor composition built from PatentsView.

Period	Non-Hisp White	API	Hispanic	Non-Hisp Black	Two+ races
1990–1999	63.8%	21.2%	4.7%	6.6%	2.7%
2000–2009	58.1%	27.6%	5.1%	5.9%	2.4%
2010–2019	51.1%	34.9%	5.8%	5.1%	2.3%
2020–2024	45.0%	41.4%	6.3%	4.5%	2.1%

Table 2: Patent-inventor surname-imputed ethnic composition by decade, $n = 3{,}121{,}166$ unique inventors with debut 1990–2024 and Census-matched surname.

The patent-side surge to API is real, large, and visible well before the trademark-side 2018 **Chinese filings shock.** API share of patent inventors rose from 21% in the 1990s to 41% in 2020–2024. The trend is smooth and starts in the 1990s — this is the immigration-driven STEM-PhD-pipeline phenomenon that Kerr (2018, *The Gift of Global Talent*, Stanford University Press) and Bernstein, Diamond, McQuade, & Pousada (2022, NBER WP) documented in administrative micro-data, not the 2018–2021 trademark cross-border filing surge. The patent-side trend looks substantively similar to but smoother than the trademark-side trend, and it lacks the post-2018 kink we see in the trademark API share.

What this means for the trademark API result.

- Some of the trademark API rise is the same real underlying process the patent data is picking up: more US-resident Asian-American filers across both systems.
- Some of the trademark API rise is the foreign-applicant surge post-2018, which has no equivalent kink in the patent data because foreign patent applicants come through different channels (PCT national-phase entries, direct-to-USPTO filings with US counsel) and don't show a comparable 2018–2020 spike.
- The trademark-corpus re-parse will let us decompose the trademark API number into US-domiciled and foreign-domiciled sub-series. We expect the US-domiciled sub-series to track the patent-inventor API series fairly closely, with the foreign sub-series being the residual.

Non-Hispanic Black share is stable across both systems. Both trademark debut filers ($\sim 11\%$ since the 1990s) and patent inventors (~ 5–7%, falling slowly) show very little movement in the Black share over thirty years. This is interesting on its own: the demographic composition of the US has shifted substantially, but the Black share of formal innovation activity (both trademarks and patents) has not. The Black share is actually lower in patents than in trademarks — consistent with the documented patent-gap literature (Cook (2014, *Journal of Economic Growth*); Bell, Chetty, Jaravel, Petkova, & Van Reenen (2019, *QJE*)).

Hispanic share rose moderately in both systems. 4.7% → 6.3% on the patent side, 6.3% → 9.4% on the trademark side. Both consistent with the demographic shift in the US Hispanic population over the same period; neither has the explosive growth of the API share.

(3) Within-firm diversification by ethnic cluster

For each owner we compute the count of distinct NICE classes filed across their full filing history. "Enclave-rooted" = an owner whose debut class is in their ethnic group's enclave list from Table 1 (e.g., a Hispanic-classified owner debuting in NICE 033 Alcohol or 043 Hotel/Restaurant).

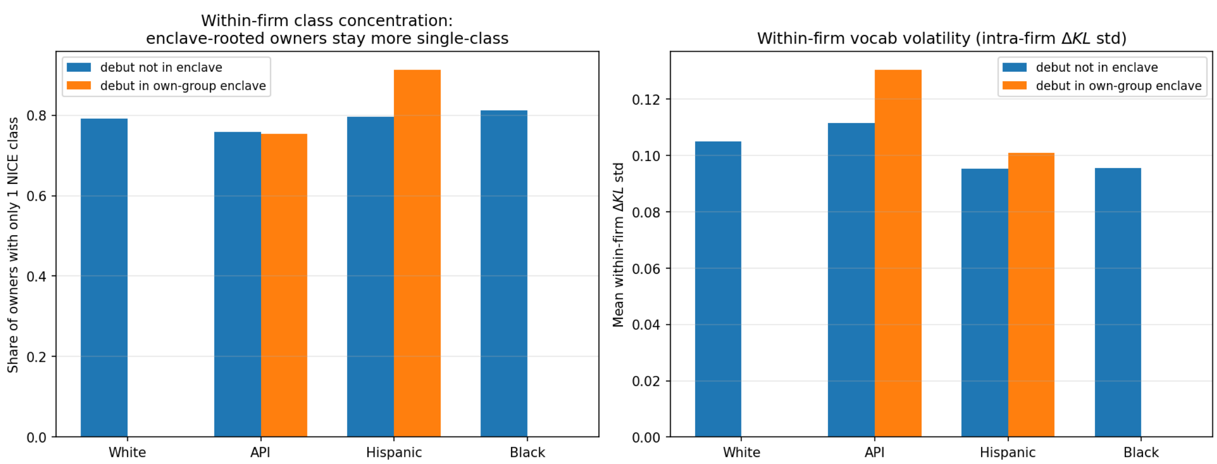

Figure 3: Left: share of owners whose entire filing history is in a single NICE class. Right: mean within-firm ΔKL standard deviation (vocab-volatility-within-firm).

Strongest single-class concentration: Hispanic enclave-rooted owners. 91.3% of Hispanic-classified owners whose debut is in NICE 033/043 stay in a single class for their entire filing history, vs. 79.6% for non-enclave Hispanic owners (**+11.7 pp**). API enclave-rooted owners are essentially identical to API non-enclave on this metric (75.4% vs. 75.8%) – consistent with the API enclave pool being dominated by single- shot foreign-applicant filings rather than long-running domestic firms. White and Black have no enclave cells.

Within-firm ΔKL volatility is similar across groups ($\approx$ 0.10 std). Enclave-rooted owners have slightly higher within-firm ΔKL volatility than non-enclave (API: 0.131 vs. 0.111; Hispanic: 0.101 vs. 0.095), but the gap is small. Within-firm vocabulary volatility is not the lever.

(4) Long-term outcomes by within-firm diversification

Restricting to owners with debut year 1995–2018 (so a 5y survival window is observable), we compare single-class, 2–3-class, and 4+-class owner cohorts on (a) registration completion rate, (b) post-registration 5y survival (proxied here by survived_5y rate over the owner's filings), and (c) SEC EDGAR listing rate.

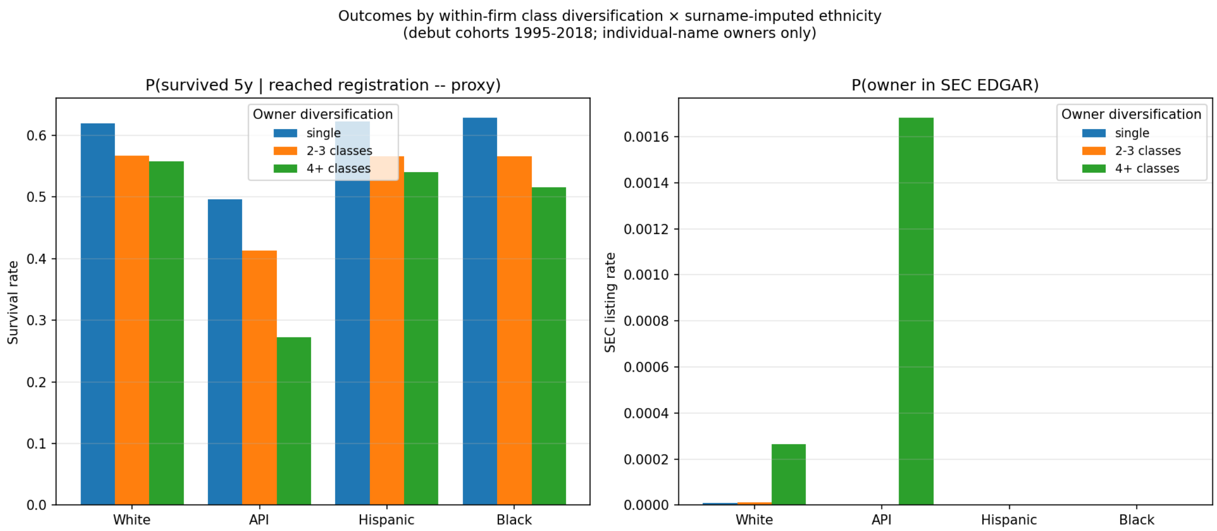

Figure 4: Outcomes by within-firm class diversification × ethnic group (individual-name owners, debut 1995–2018).

Diversification tier	n owners	P(reached)	P(survived 5y)	P(SEC listed)
Single NICE class (1)	1,967,969	53.0%	52.8%	0.09%
2–3 classes	661,126	49.3%	47.7%	0.38%
4+ classes	198,426	47.6%	43.8%	2.07%

Table 3: Pooled across all owners 1995–2018. Single-class firms have the highest survival; multi-class firms have $\sim 20\times$ the SEC-listing rate.

The trade-off is sharp: stability vs. scale. Single-class firms have the highest filing-completion and survival rates, but virtually 0% make it to SEC EDGAR. 4+-class firms have the lowest survival per filing but 2.1% reach the public market — about 23× the single-class rate. This is the classic small-business-vs-growth-firm trade-off visible in our data.

For Hispanic and Black single-class firms specifically, survival is highest in our sample. Hispanic single-class: 62.5% reached registration, 62.2% survived 5y. Black single-class: 63.2% reached, 62.9% survived. Both exceed the pooled and White single-class numbers. Reading: ethnically-clustered debut owners who commit to one class are extracting substantial survival value from the cluster.

API single-class survival is anomalously low ($\sim 50\%$). Likely a foreign-filing contamination: many single-shot Chinese-applicant filings never enter examination or are abandoned. The re-parse will strip these and recompute.

(5) Plumbing vs. computing: the "less-sexy" sector test

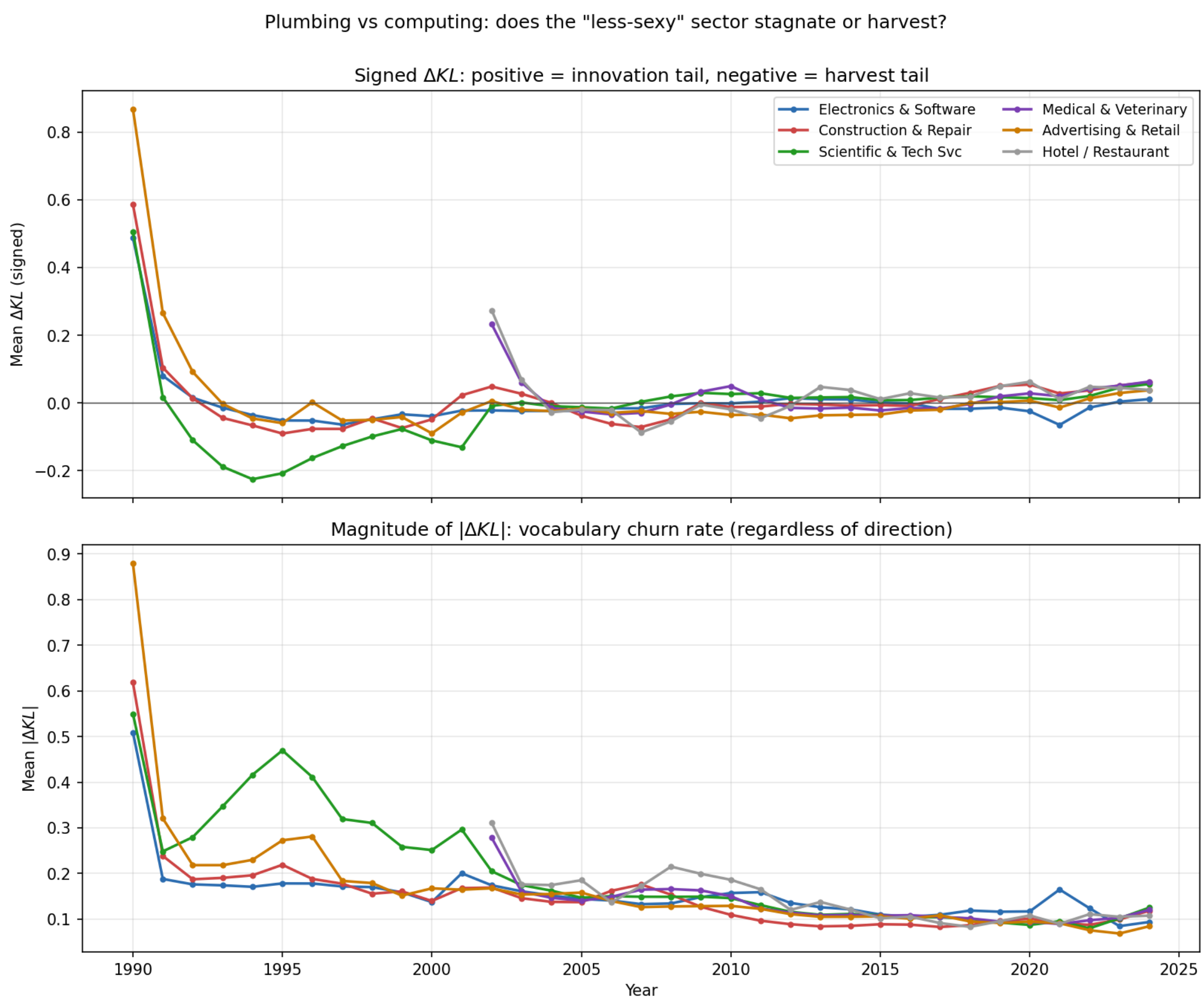

Figure 5: Mean ΔKL (top, signed) and $|\Delta KL|$ (bottom, magnitude/churn) for six NICE classes, 1990–2024.

Construction & Repair is on the innovation tail, harder than Electronics in 2020–2024. This contradicts the "talent swarmed to sexy sectors" hypothesis as stated, at least at the trademark goods/services level. NICE 037 (construction/repair, includes plumbing services) shows the *highest* positive ΔKL in 2020–2024 (+0.047). Plausible mechanism: solar installation, EV charging

NICE class	1995–1999 signed ΔKL	2010–2014	2020–2024
009 Electronics & Software	−0.050	+0.008	−0.019
037 Construction & Repair	−0.073	−0.008	**+0.047**
042 Scientific & Tech Svc	−0.131	+0.020	+0.029
044 Medical & Veterinary	–	+0.001	+0.040
043 Hotel/Restaurant	–	+0.006	+0.040

Table 4: Three-period signed ΔKL for six NICE classes. Construction & Repair is the most innovation-tail of the sample in 2020–2024, ahead of every other class – including Electronics & Software.

infrastructure, smart-home electrification, sustainable building, ADU construction – a wave of genuinely new vocabulary entered the trade. Electronics & Software (ΔKL between -0.02 and $+0.01$, basically flat) has settled into stable vocabulary because everyone already shares the same terms. The magnitude $|\Delta KL|$ is comparable across classes (~ 0.10 in 2020–24); no class is stagnant in the zero-vocabulary-churn sense.

(6) Patent-side: inventor-surname HHI by CPC class over decades

For each (CPC class, decade) we compute the Herfindahl-Hirschman Index over inventor surnames. Rising HHI = the same surnames keep showing up; falling HHI = diversifying.

CPC class	HHI 1990s	HHI 2020s	Description
Rising surname concentration (top 5)			
G09	0.00061	0.00691	Educating; cryptography; display; advertising
H10	0.00090	0.00715	Semiconductor devices (new section)
D06	0.00073	0.00685	Treatment of textiles; laundering
B09	0.00121	0.00496	Disposal of waste; reclamation
G11	0.00076	0.00424	Information storage / memory
Falling surname concentration (top 5, diversification)			
G16	0.00212	0.00062	Information and communication tech
F03	0.00178	0.00114	Machines/engines for liquids/gases
B64	0.00089	0.00038	Aircraft, aviation, cosmonautics
F41	0.00113	0.00068	Weapons
A24	0.00211	0.00168	Tobacco; smokers' articles

Table 5: All HHI values are small (the top class has $\sim$ 1/145 effective surnames in its tail), so absolute magnitudes are modest; the directional story is the change.

The rising-HHI classes are heavily hardware-manufacturing classes. The falling-HHI classes are ICT and mature mechanical/aerospace. This is a guild signal in the narrow sense: some classes are seeing growing surname concentration, others are seeing diversification.

The honest read is ethnic concentration, not family inheritance. Chinese surnames are roughly 20× less diverse than Western surnames – the top 100 Chinese surnames cover $\sim$ 85% of the Han Chinese population, vs. top 100 Western surnames covering $\sim$ 15% of US Whites. So if the inventor pool in a CPC class shifts toward Chinese names, the surname HHI rises mechanically, even without any actual family-inheritance dynamic. The rising-HHI CPC classes (G09, H10, D06, G11, G02, H01) are precisely the semiconductor / display / memory / optics / basic-electrical classes where

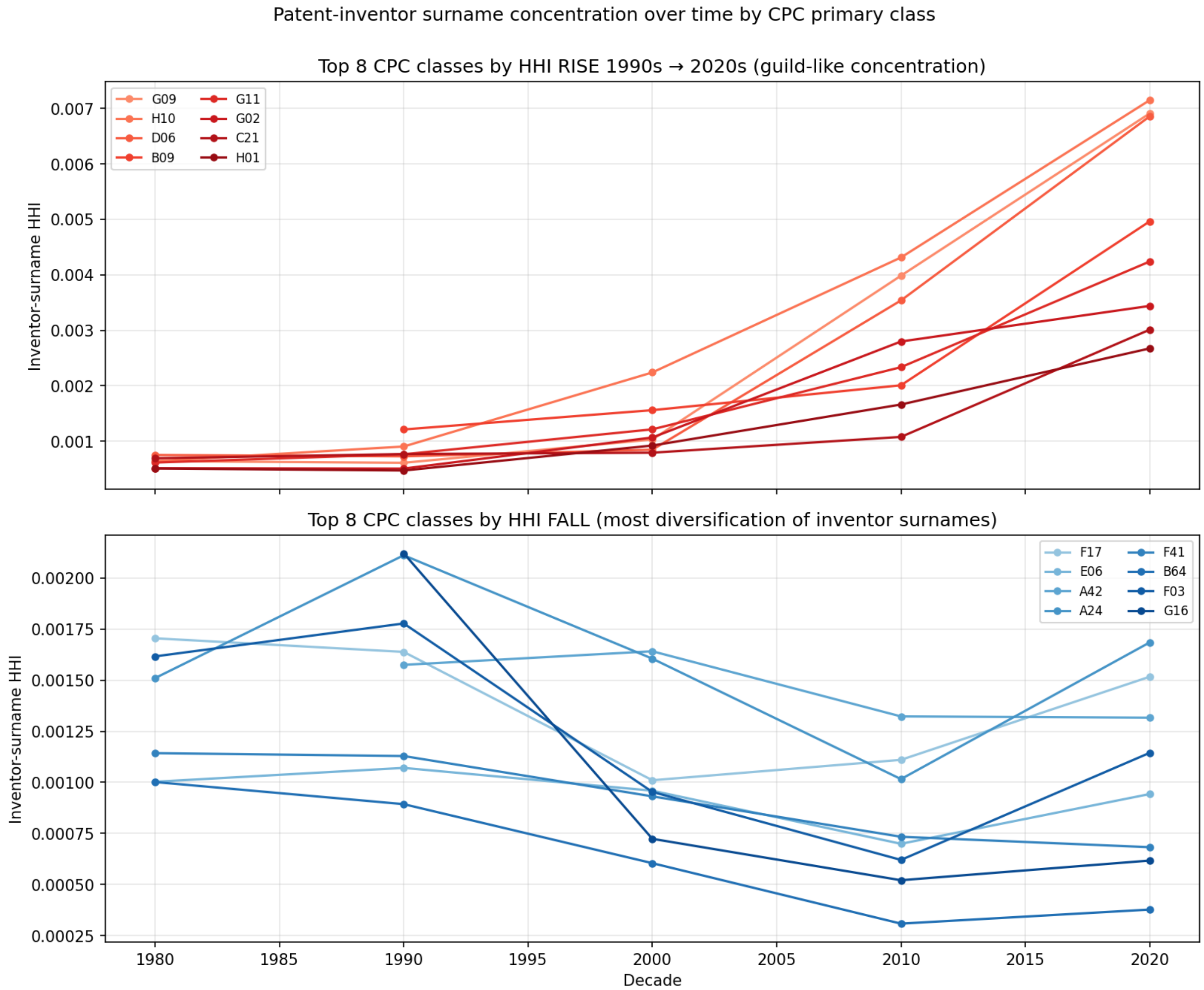

Figure 6: Top: eight CPC classes whose inventor-surname HHI rose most from the 1990s to the 2020s. Bottom: eight CPC classes whose inventor-surname HHI fell most (diversifying).

Chinese-resident inventor share has been documented to rise sharply since 2010 (Hicks, 2024, *Stanford Law Review*). So the pattern is more *ethnic-group concentration in technical sub-fields* than *family-line inheritance across generations*. To get to a true guild test we'd need to disambiguate Chinese names across generations (which the disambiguation in PatentsView does at the inventor level but cannot do at the family level), and we'd need a US-resident inventor restriction (which requires the location_id→country crosswalk; doable as a follow-up).

The falling-HHI CPC classes – G16 in particular, which dropped from 0.00212 in the 1990s to 0.00062 in the 2020s (-71%) – show the opposite pattern: surname diversification consistent with broad workforce expansion. G16 is information and communication technology, where the workforce went from a small concentrated pool ($\sim$ 800 inventors with a few dominant surnames) to a much larger and more diverse pool. This is the Hsieh-Hurst-Jones-Klenow (2019, *Econometrica*) talent reallocation story playing out in patent micro-data.

What's next

- Full USPTO XML re-parse for state+country ($\sim$ 30–60 minutes, currently $\sim$ 30% through). Once it lands: rebuild the trademark debut composition stripping foreign-domiciled filings, and recompute the ethnic-cluster ΔKL table to remove the foreign-applicant confound on API enclaves. Same applies to the patent-side surname HHI – restricting to US-resident inventors will let us separate "ethnic concentration of US inventors" from "rising Chinese-resident filing volume."
- PatentsView location join (`g_location_disambiguated.tsv`) to attach inventor country/state. Re-run the HHI analysis on US-only inventors; predict that several of the rising-HHI CPC classes will look much flatter once foreign inventors are stripped.
- Within-firm ΔKL trajectory analysis: do firms drift toward or away from their starting ΔKL over subsequent filings? Is the drift ethnically systematic?

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